Shashwat Jaguri

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Education

University of Michigan — Ann Arbor, MI
B.S.E. in Computer Engineering, GPA: 3.62 — *Expected Grad: May 2027*
M.S.E in Electrical & Computer Engineering (SUGS Program) — *Expected Grad: May 2028*

Relevant Courses: Computer Architecture (EECS 470), VLSI Design (EECS 427), Digital Integrated Circuits, Digital Logic Design, Programming & Data Structures

Experience

Undergraduate Research Assistant — May 2026 – September 2026
M-Bits Lab, University of Michigan | Advisor: Prof. Reetuparna Das — *Ann Arbor, MI*

- Built a custom C++ gem5 SimObject modeling Intel AMX that matches measured hardware performance within 6% while remaining fully behaviorally correct, establishing a cycle-accurate baseline for evaluating memory-side inference accelerators.
- Wrote optimized GEMM kernels in C using AMX intrinsics that outperformed Intel's oneDNN by a geomean of 7.8%, used to validate simulation accuracy against real workloads.
- Profiled Sapphire Rapids CPUs with targeted microbenchmarks to measure AMX latency and throughput, then analyzed the measured data to calibrate and validate a performance simulator.

Instructional Grader for EECS 470 — August 2026 – Present
CSE Department, University of Michigan — *Ann Arbor, MI*

- Grade homework, labs, and exams on out-of-order processor design, memory hierarchies, and verification.

Club President — April 2026 – Present
RISC-M, University Student Organization — *Ann Arbor, MI*

- Co-founded UofM's first VLSI/computer architecture club; grew it to 40+ members and secured sponsorships.

Projects

Superscalar Out-of-Order RISC Processor and Toolchain — March 2026 – May 2026

- Designed and synthesized a 2-way superscalar, 5-stage out-of-order processor on a 4-person team, reducing average time per instruction from 30 ns to 12.14 ns over the 3-stage in-order baseline.
- Engineered a micro-op queue splitting memory instructions into address-generation and memory-access operations, decoupling execution from the front-end.
- Integrated a gshare branch predictor, 2-way banked BTB, and set-associative I-cache to reduce front-end stalls.
- Verified the micro-op queue with standalone SystemVerilog testbenches targeting structural and handshake hazards; validated full-core correctness via architectural state trace diffing and waveform debugging.

Full-Custom 16-bit RISC Processor & Edge Accelerator — October 2026 – Present

- Designing a 16-bit RISC CPU and edge navigation accelerator SoC in TSMC 65nm on a 6-person team, with a dual-clock architecture and memory-mapped CSR control.
- Completed full-custom layout of the register file and ALU with a hybrid Kogge-Stone adder, part of a fully hand-laid-out CPU datapath (APR used only for control logic); DRC/LVS-clean in Mentor Calibre.
- Implemented layout of a 16-bit radix-4 logarithmic shifter, reused in the accelerator's iterative CORDIC unit.
- Currently designing the 2b-weight/2b-activation SRAM digital compute-in-memory (DCIM) macro: weight-stationary 6T bitcells, NAND2-based per-lane multiplication, and a reduction tree with weighted shift-add.

Neural Guitar Effects Pedal — September 2026 – Present

- Building a guitar effects pedal on an XMC4700 MCU (3-person team): ADC/DAC audio, encoders, RGB LEDs, LCD.
- Brought up the full audio chain from guitar input through ADC, on-chip DSP effects (e.g., gain), and DAC output.
- Currently developing a WaveNet neural amp model on an FPGA, streamed from the MCU over SPI DMA with CDC.

Technical Skills

Languages: Python, C, C++, SystemVerilog, Verilog, MATLAB, Assembly (ARMv8, x86)
Tools & Technologies: Linux, Git, gem5, PyTorch, Virtuoso, Spectre, HSPICE, Synopsys, Verdi, Quartus, Altium, STM32, XMC4700